

# Gavin Abrigo

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MLOps, Data Engineering, AI Engineering, Project Management, Agentic Workflows

## EDUCATION

### University of California Merced

*Bachelor of Science in Computer Science & Engineering*  
*Minor in Management Analytics and Decision Making*

Merced, CA

May 2027

## TECHNICAL SKILLS

**Languages** — TypeScript, JavaScript, GoLang, Java, Python, C/C++, HTML/CSS, LaTeX, MIPS Assembly, SQL, PostgreSQL, MATLAB, SPARQL,

**Technologies** — Shell, React.js, FastAPI, REST APIs, Git, GitHub, Node.js, Next.js, Vite, StreamLit, Nvidia NIM, Tesseract, OpenCV, Docker, PyTorch, Pandas, Excel, SQLite, CI/CD, Git, MongoDB, Flask, JSON

## EXPERIENCE

### Software Engineer Intern

January 2026 - Present

*BART - Software Engineering Capstone*

Merced, CA

- Automatically classify infrastructure incident images submitted via web/mobile app, utilizing structured metadata, and route to appropriate maintenance teams
- Developing image embedding and similarity detection workflows (Google Gemini) to de-duplicate submissions with postgres vector embeddings, extract contextual location data, and reduce manual review overhead.

### Undergraduate Researcher Intern

September 2025 - Present

*LLM Knowledge Graph Retrofitter (KGR) - Prof. Pandey*

Merced, CA

- Improve factual accuracy to reduce hallucinations in NLP models. Building a Python-based Knowledge Graph Retrofitting (KGR) and Retrieval-Augmented Generation (RAG) pipeline using Wikidata and GraphRAG.
- Fine-tuning entity linking systems and evaluating LLM verification performance with NLP metrics such as exact match and F1 on benchmark datasets.

### Software Engineer Intern (backend)

January 2025 - March 2025

*OculusIO (NFT Automation Software Startup)*

Remote

- Developed Bash and TypeScript CLI automation tools to streamline multi-language builds, integrating Go protobuf generation and client compilation—cutting setup time by 65% and build errors by 40%.
- Implemented cross-platform environment validation scripts for Windows, macOS, and Linux, standardizing workflows and improving developer onboarding efficiency across a 6-person team.

## PROJECTS

### SMART BART | Software Engineering Capstone | Python, Next.js, FASTAPI, AI fullstack | May 2026

- Image-based incident reporting system for Bay Area Rapid Transit (BART) that auto-classifies submitted maintenance photos and routes to correct department
- Gives maintenance workers a unified workflow for turning scattered photo reports into organized maintenance intelligence.
- Used vector embeddings, SentenceTransformers, and cosine similarity to cluster duplicate incident photos, so BART staff see one grouped report

### NYC Taxi Trip Duration Regression and Digit Classification | Python, Scikit-learn, Pandas | Oct 2025 - Dec 2025

- Predicting total ride duration of taxi trips in New York City, Fraudulent transaction classification, and MNIST multidigit classification
- Building and evaluating classification and regression models across MNIST, Taxi Trip Duration, and Credit Card Fraud datasets.
- Implemented logistic regression with hyperparameter tuning and XGBoost with 5-fold stratified cross-validation to predict total NYC taxi trip durations and classify fraudulent credit card transactions.